

# Analogy as the core of cognition (?)<sup>1</sup>

---

Nishit Singh <sup>2</sup>

June 2026

Birla Institute of Technology and Science (BITS), Pilani

Project update, 14th August 2026

---

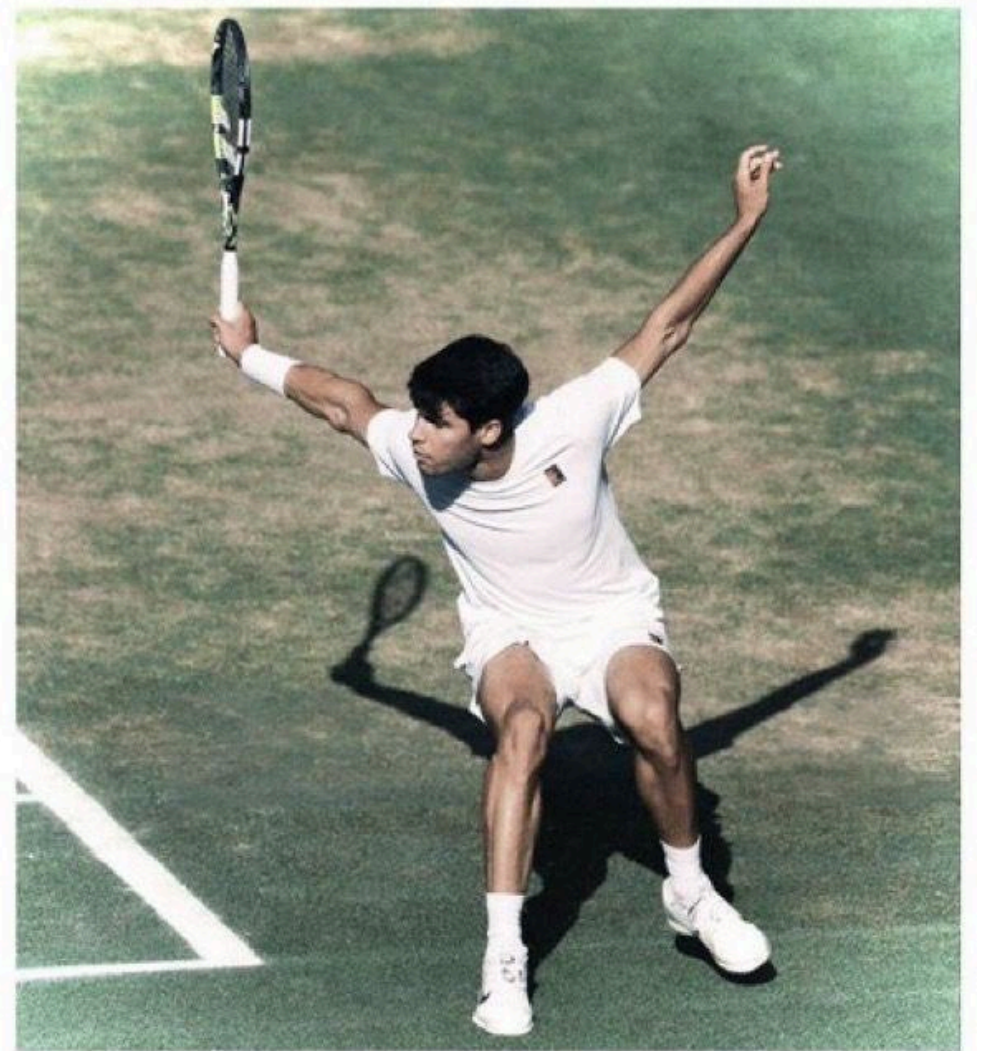
[1] Douglas Hofstadter, [\*Stanford Presidential Lecture\*](#), 2009

[2] [Link](#) to personal webpage

# How hard is it for a world class ballerina to win a grand slam?



??



# Definition(s)

---

**Analogy is the mapping of knowledge from a base domain to a target such that a system of relations holding among the base objects also holds among the target objects, with relational structure mattering far more than shared object attributes. [Gentner, D. \(1983\)](#)**

Analogical mapping between a source and target is governed by three interacting constraints: structural [isomorphism], semantic [similarity], and pragmatic [purpose], satisfied in parallel rather than by structure alone. *Holyoak & Thagard (1989)*

Analogy-making is the perception of common essence between two things, treated not as a specialized reasoning tool but as the basic mechanism underlying all concepts and thought. *Hofstadter (2001)*

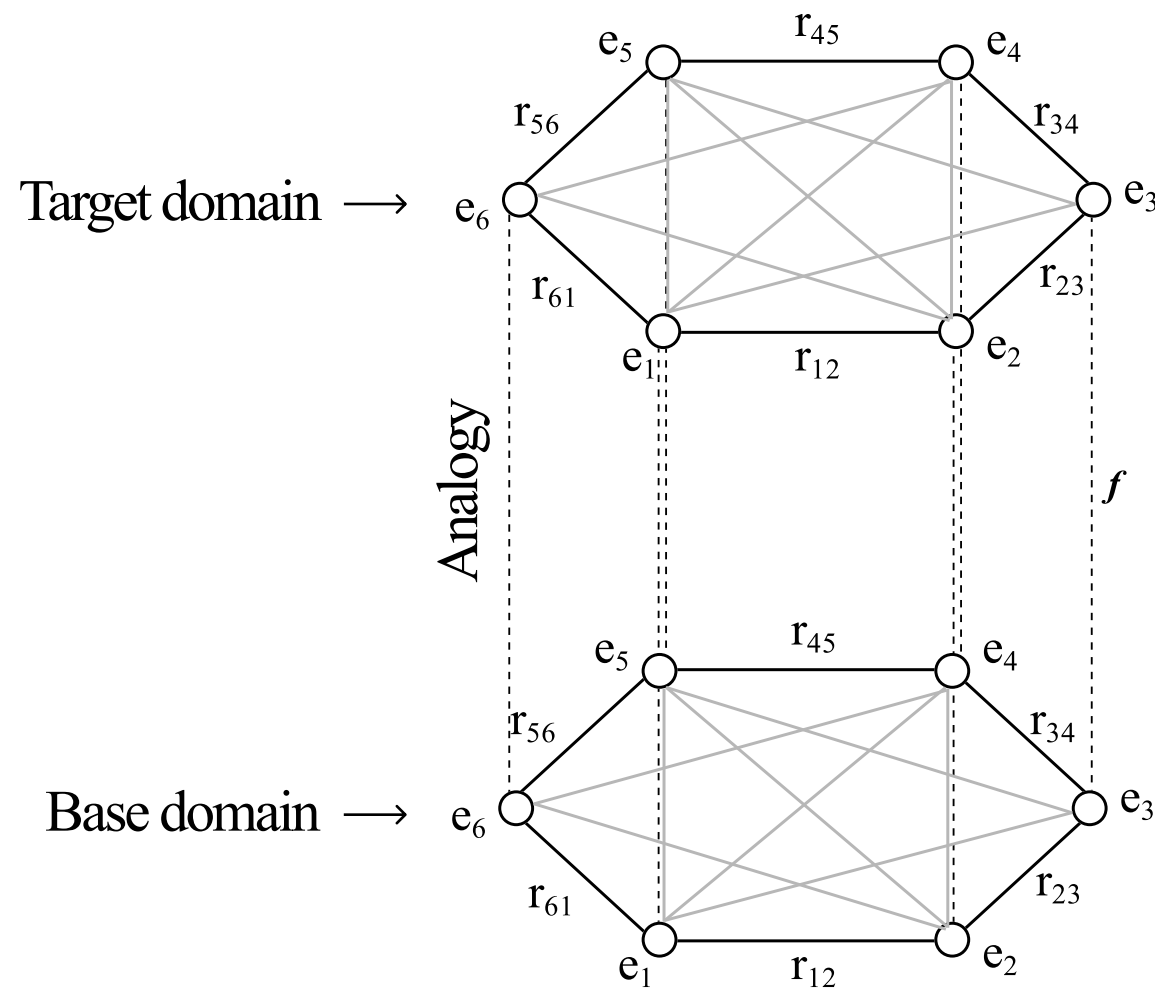
Analogy is the use of a solution to one problem to generate a solution to a structurally similar problem, the empirical basis for studying spontaneous vs. cued transfer.

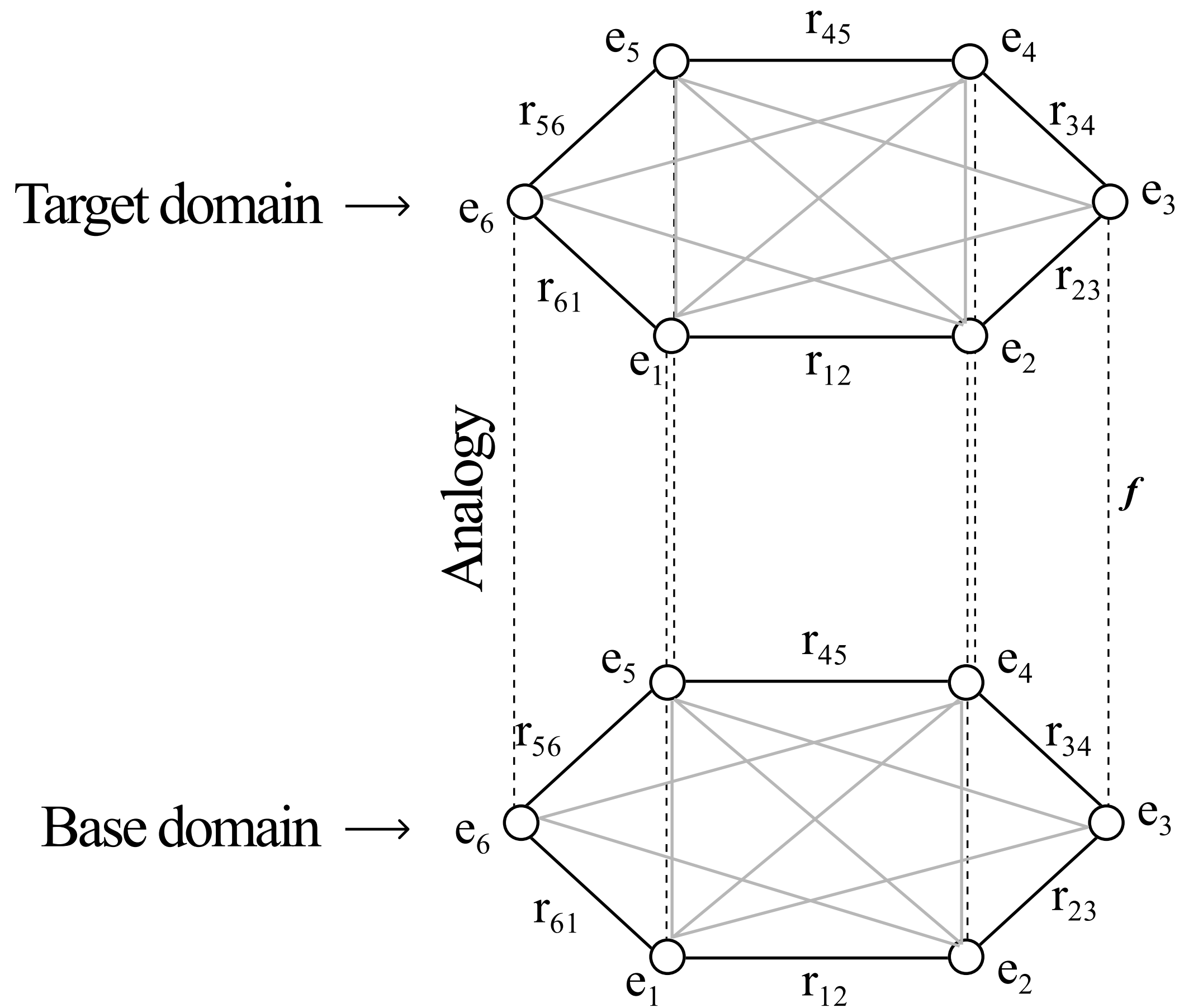
Analogy is a structure-sensitive process of access and mapping over distributed representations, combining the flexibility of a connectionist system and the structure sensitivity of a symbolic system.

# Definition

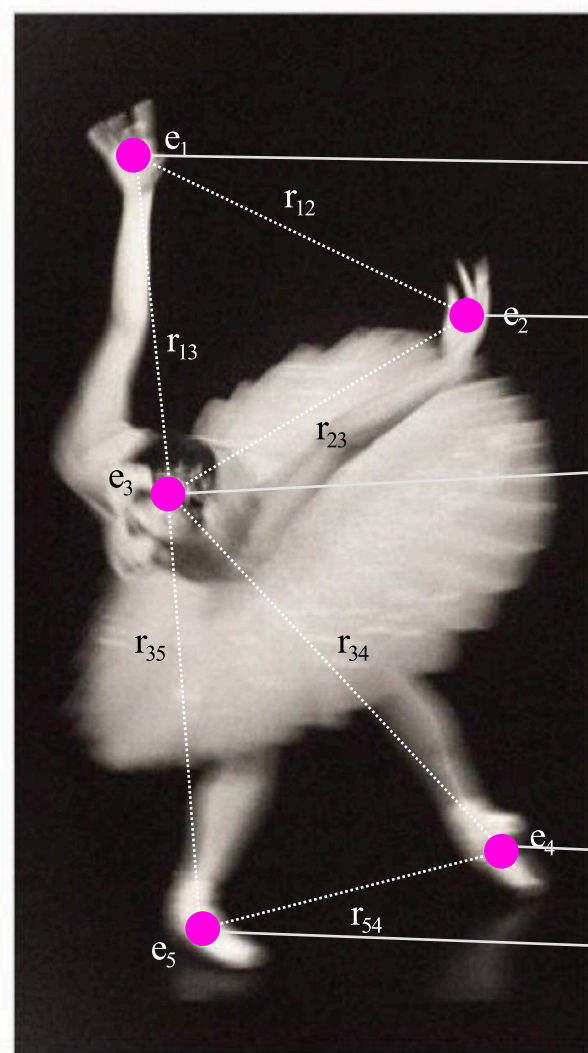
---

Analogy is the mapping of knowledge from a base domain to a target such that a system of relations holding among the base objects also holds among the target objects, with relational structure mattering far more than shared object attributes. [\*Gentner, D. \(1983\)\*](#)





# Definition



Base Domain

ANALOGY LINE

ANALOGY LINE

ANALOGY LINE

ANALOGY LINE

ANALOGY LINE



Target Domain

For this setup, it is prudent to express analogy as directly proportional to “how parallel” the analogy lines are. This definition is not necessarily general; but it works sufficiently here.

# Related work

---

## Emergent Analogical Reasoning in Transformers



*Gouki Minegishi, Jingyuan Feng, Hiroki Furuta, Takeshi Kojima, Yusuke Iwasawa, Yutaka Matsuo*

 Published: 01 May 2026, Last Modified: 25 Jun 2026  ICML 2026 spotlight  Everyone  Revisions  BibTeX  CC BY 4.0

**TL;DR:** We analyze how analogy is implemented inside Transformers and show that the same mechanism extends to pretrained LLMs.

**Abstract:**

Analogy is a central faculty of human intelligence, enabling abstract patterns discovered in one domain to be applied to another. However, the mechanisms underlying analogical reasoning in Transformers remain poorly understood. In this work, inspired by the notion of functors in category theory, we formalize analogical reasoning as the inference of correspondences between entities across categories. Based on this formulation, we introduce synthetic tasks that evaluate the emergence of analogical reasoning under controlled settings. We find that the emergence of analogical reasoning is highly sensitive to data characteristics, optimization choices, and model scale. Through mechanistic analysis, we show that analogical reasoning in Transformers decomposes into two key components: (1) geometric alignment of relational structure in the embedding space, and (2) the application of a functor within the Transformer. These mechanisms enable models to transfer relational structure from one category to another, realizing analogy. Finally, we quantify these effects and find that the same trends are observed in pretrained LLMs. In doing so, we move analogy from an abstract cognitive notion to a concrete, mechanistically grounded phenomenon in modern neural networks.

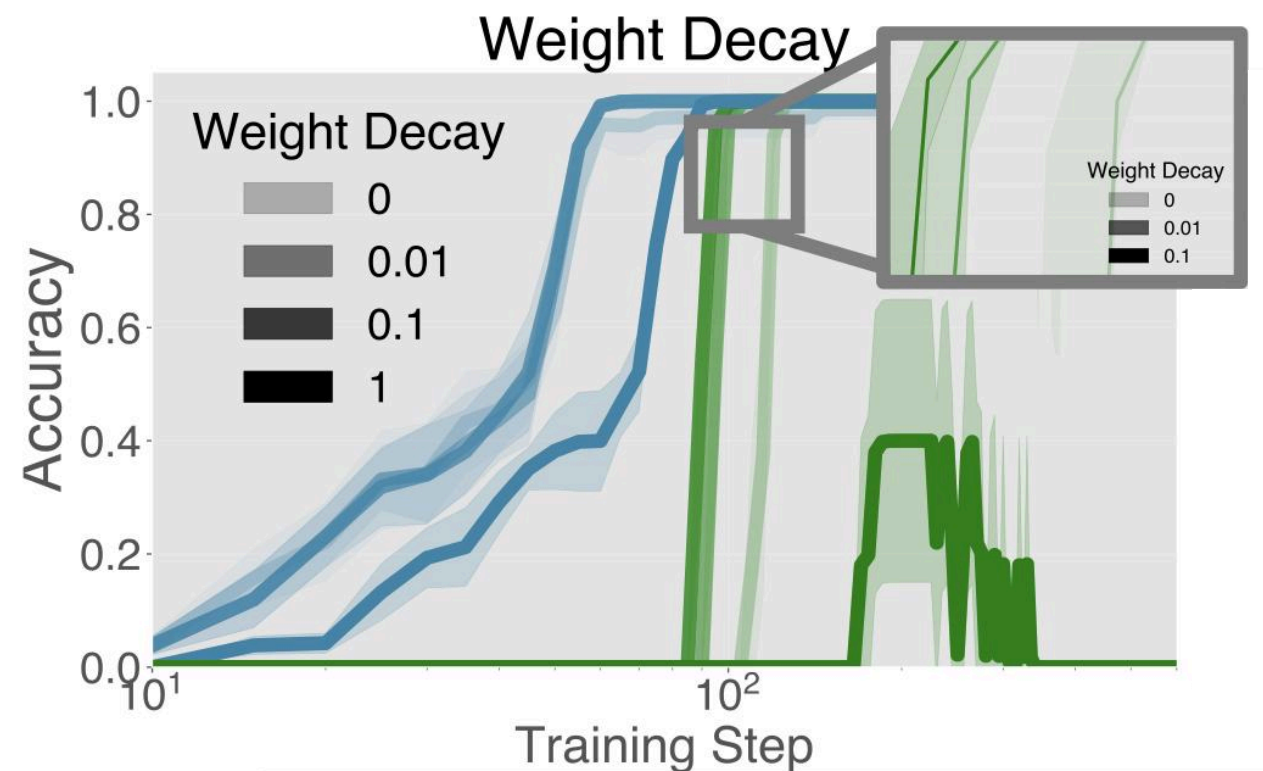
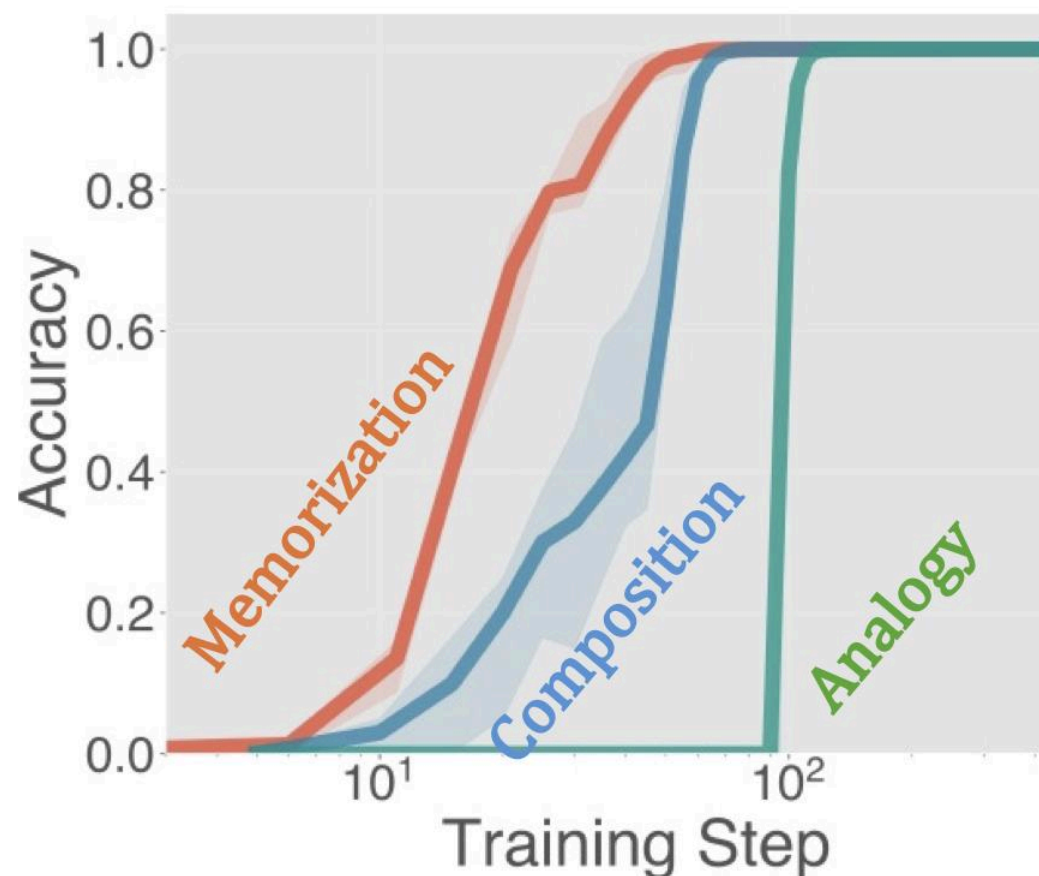
[Spotlight paper, ICML 2026.](#)

# What we already know

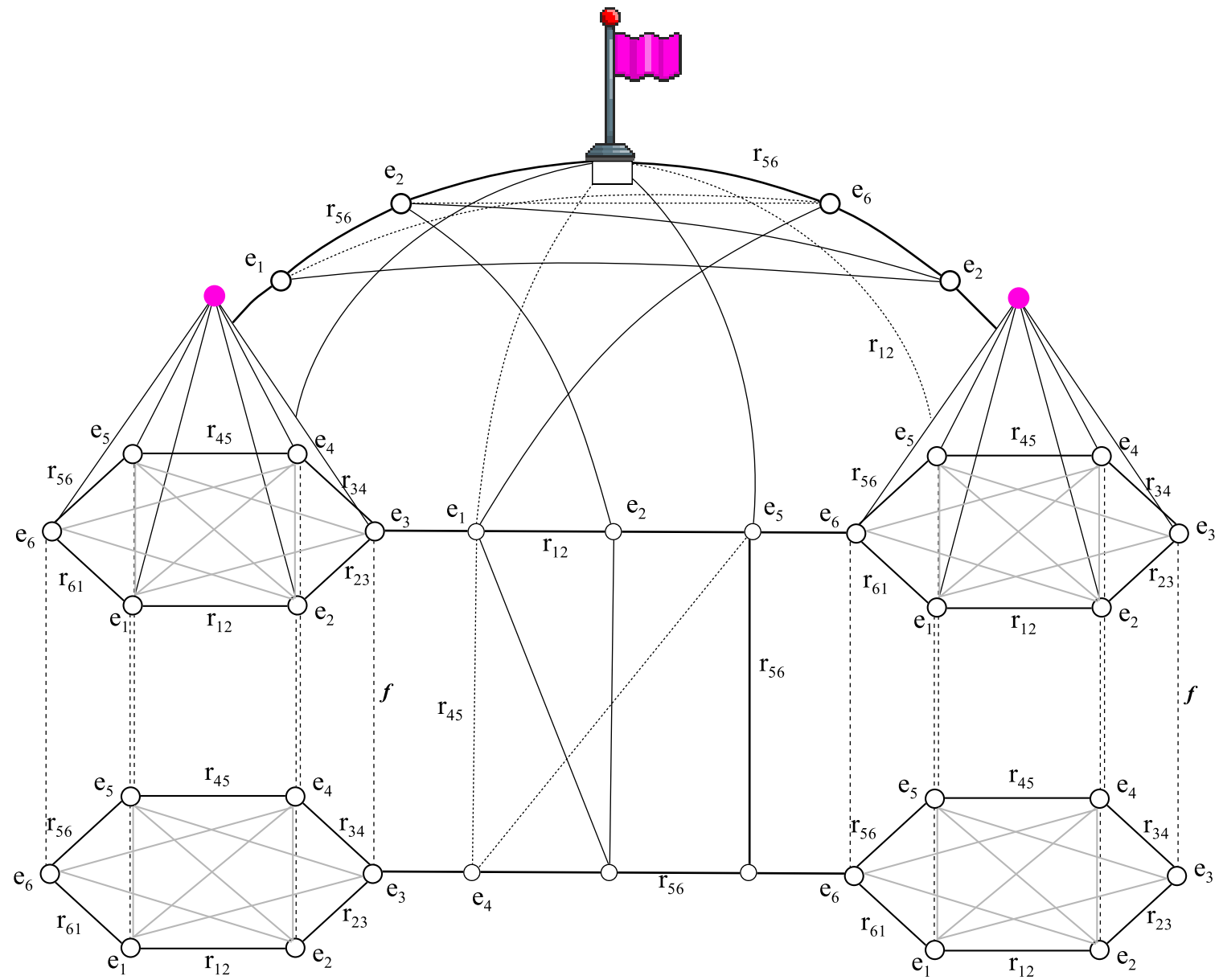
$$(e_s, r(e_s, e_t), e_t) \in D_{\text{atomic}}$$

$$(e_s, r(e_s, e_i), r(e_i, e_t), e_t) \in D_{\text{comp}}$$

$$(e_s, f, \mathcal{F}(e_s)) \in D_{\text{analogical}},$$

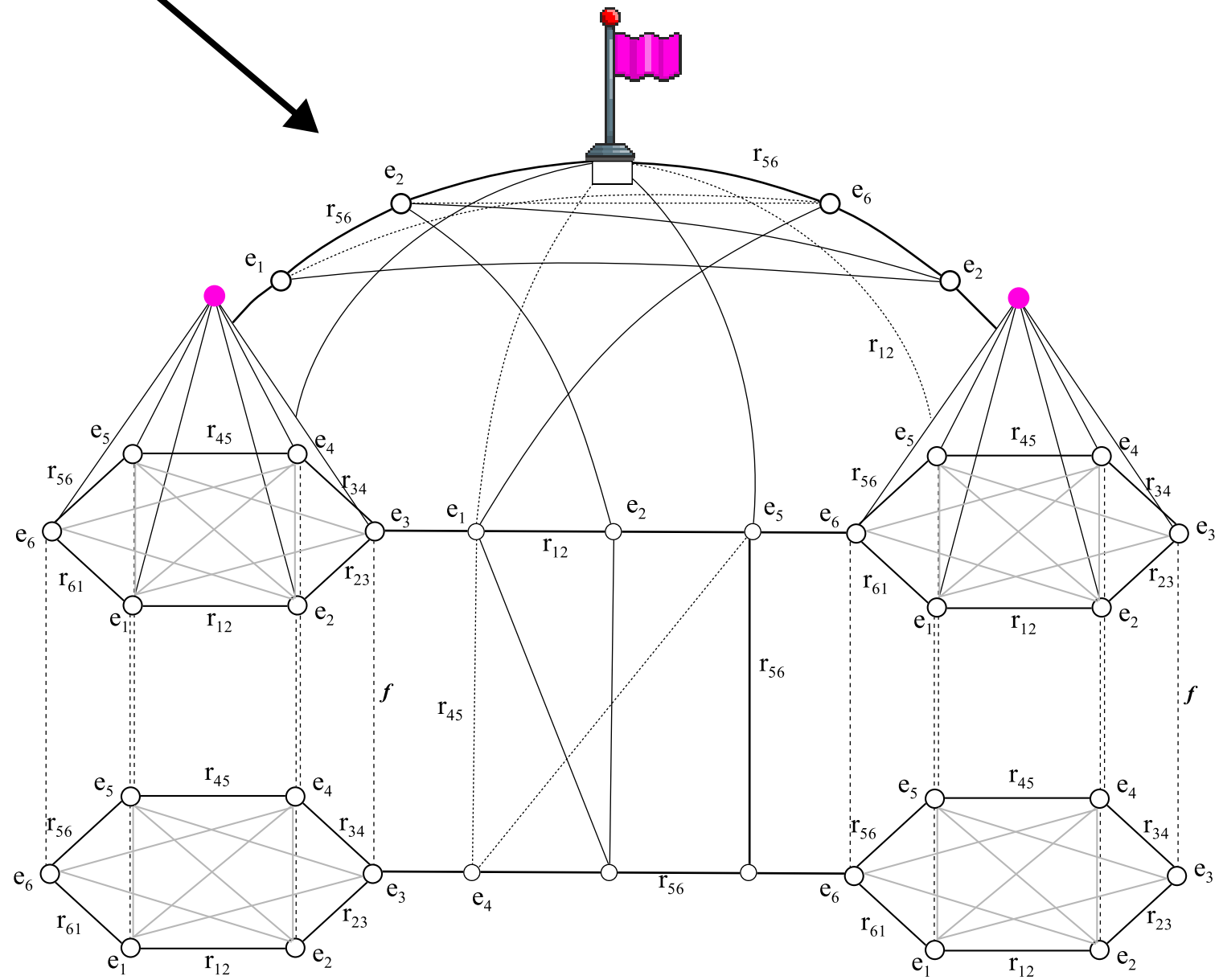


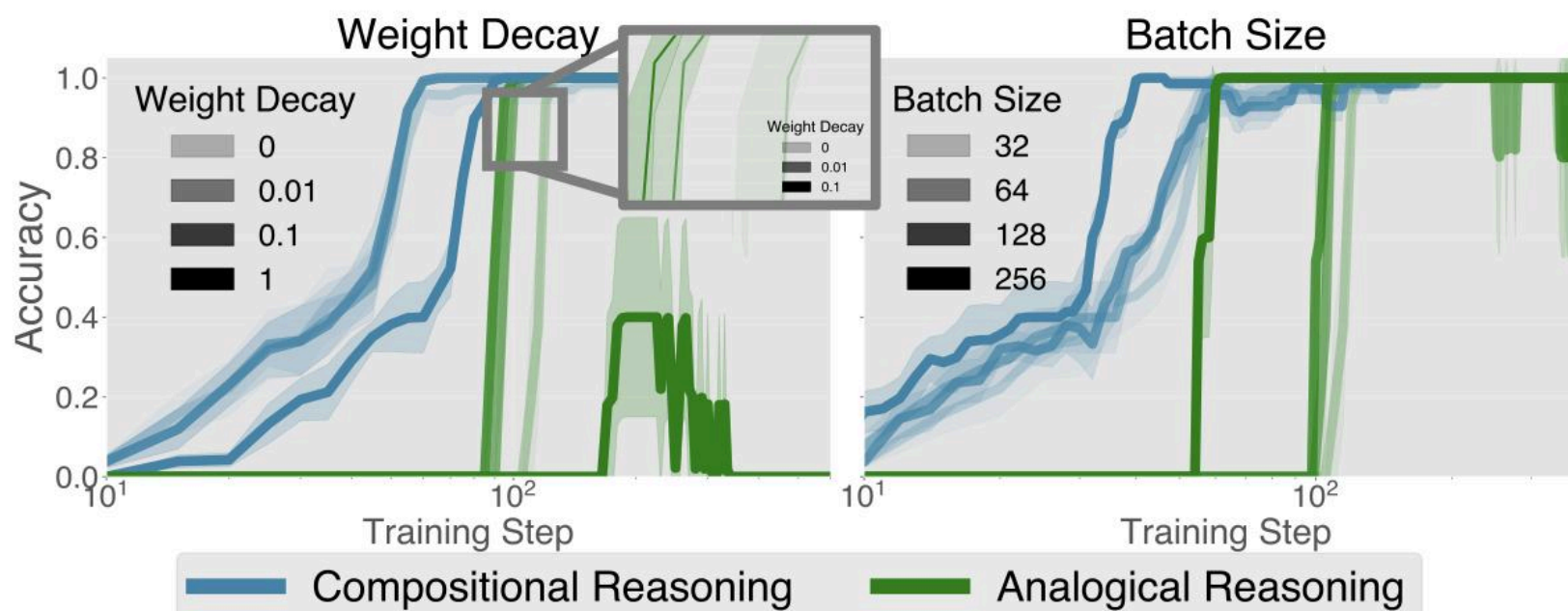
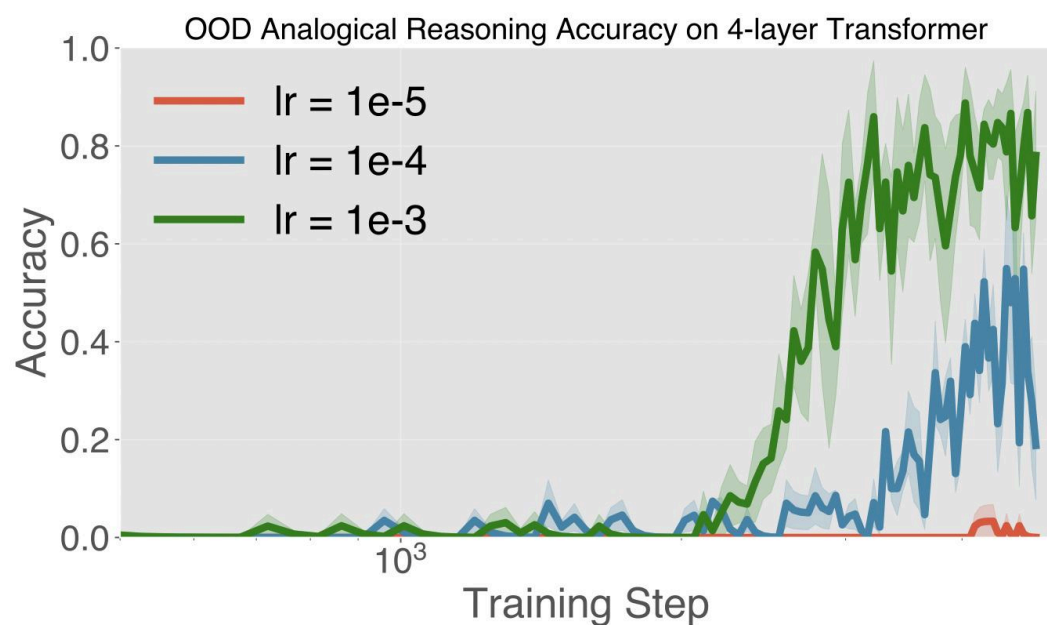
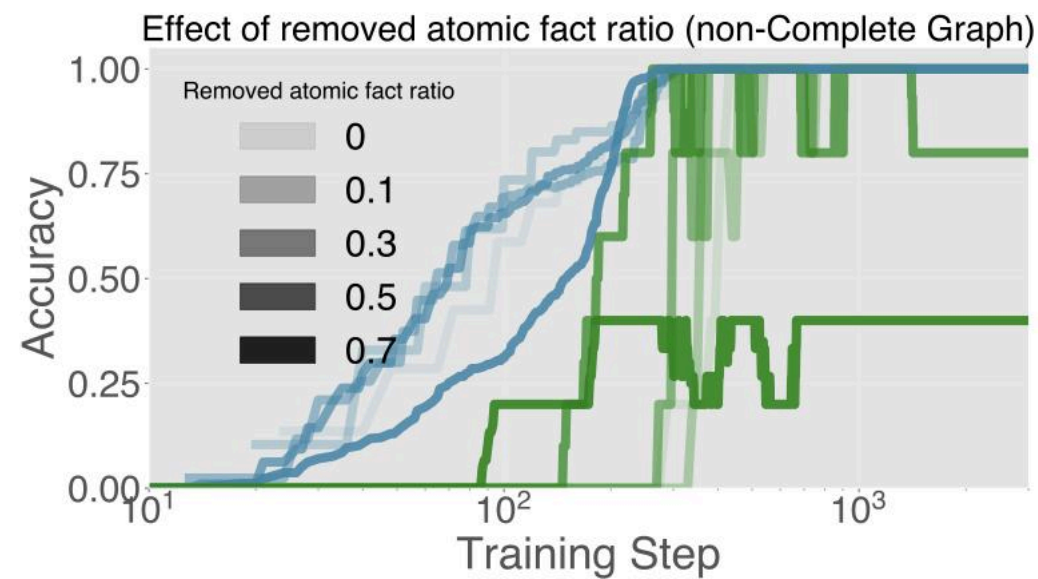
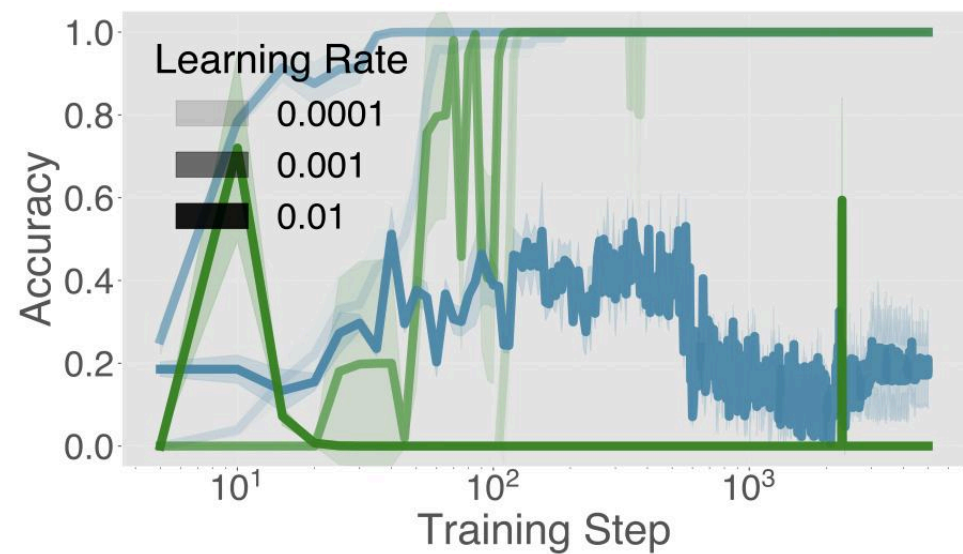
Attacking the **category theory** fortress.



Why is it fickle?

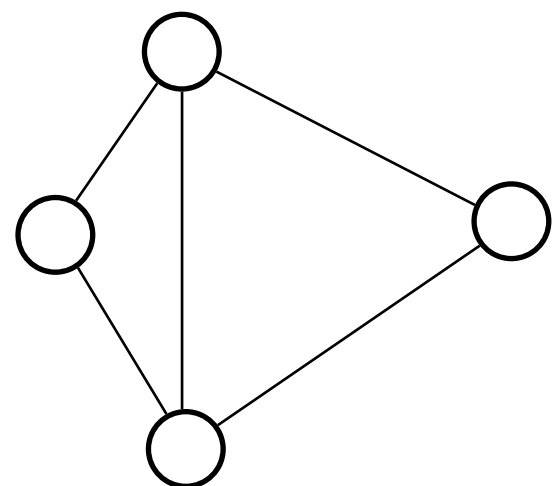
[ RCAs ]



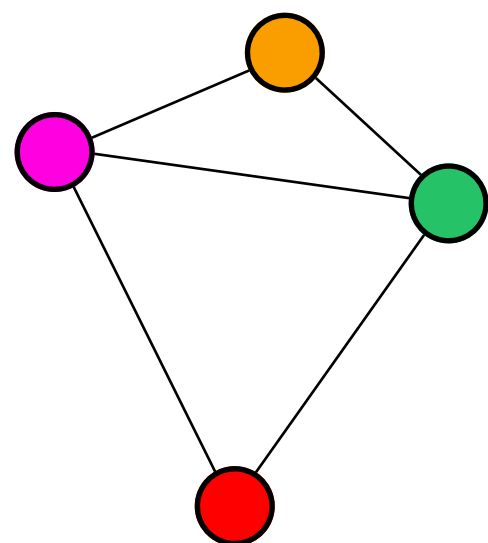
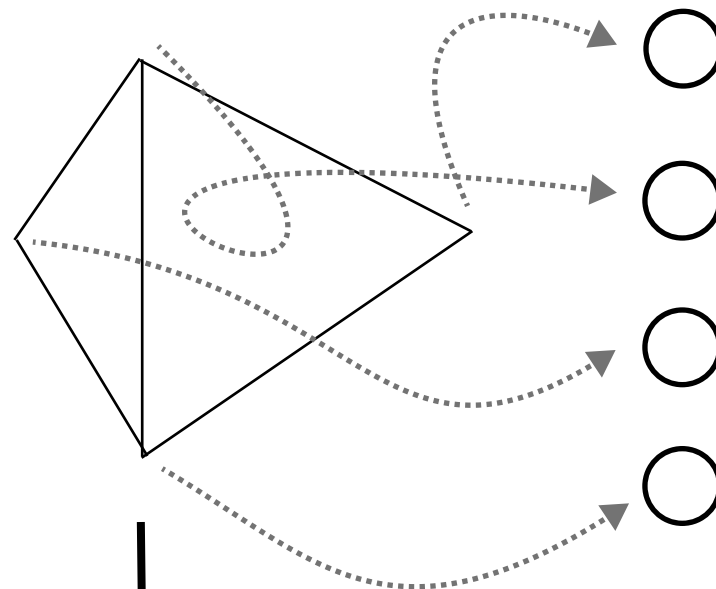




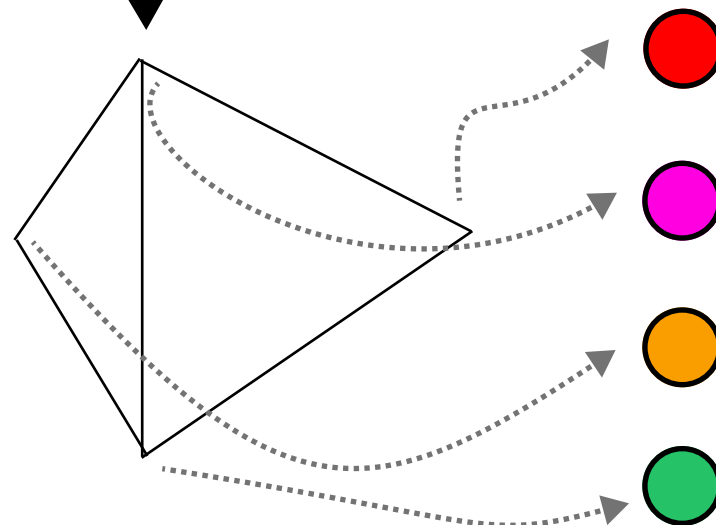
POV: how **Relational Cross Attention (RCA)**  
transformers look at you when they separate the  
relational and contextual halves of the problem,  
and learn relationships ripped off of their  
contextual role *somehow increasing accuracy*

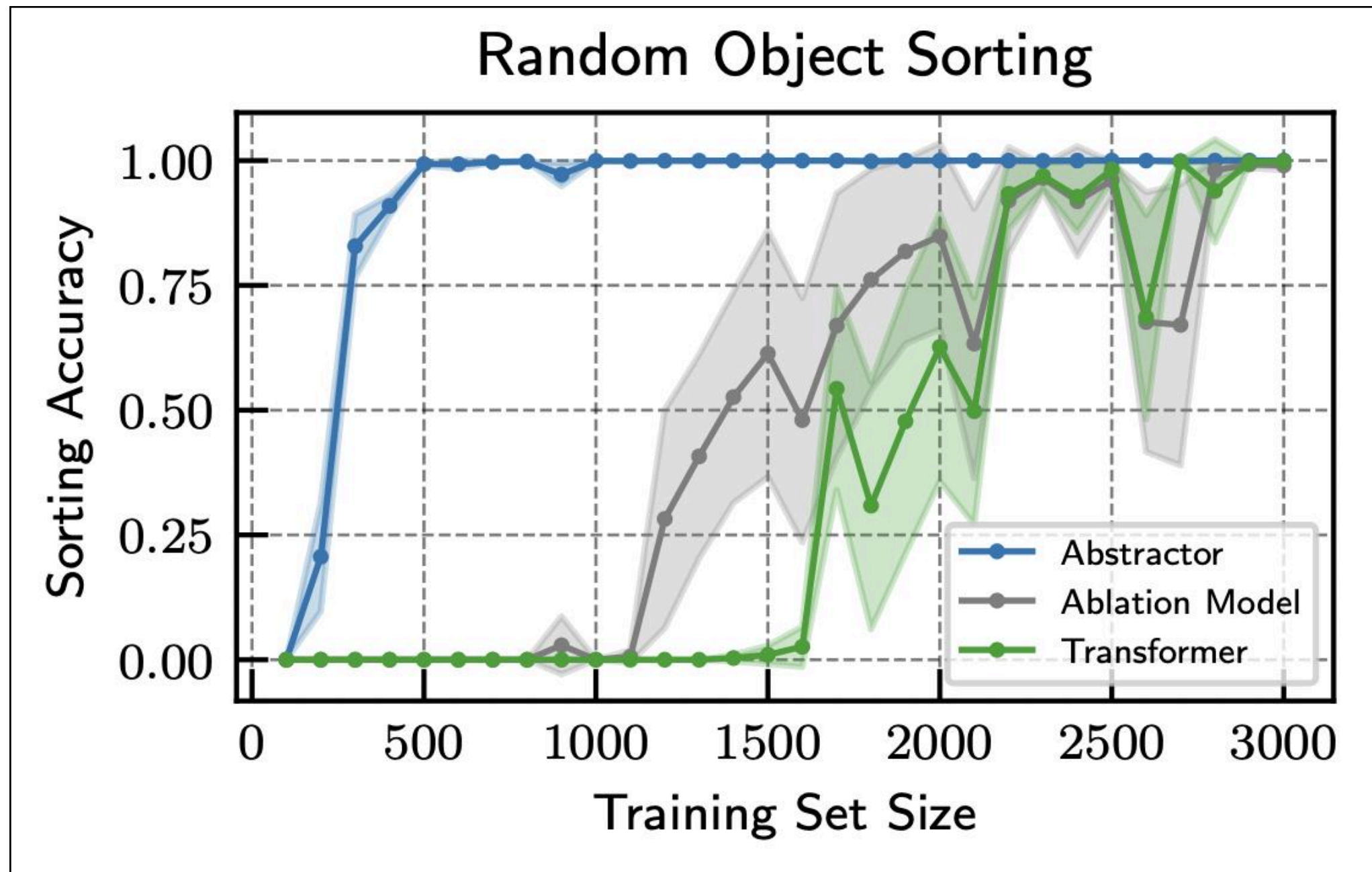


**RCAT**



**RCAT**

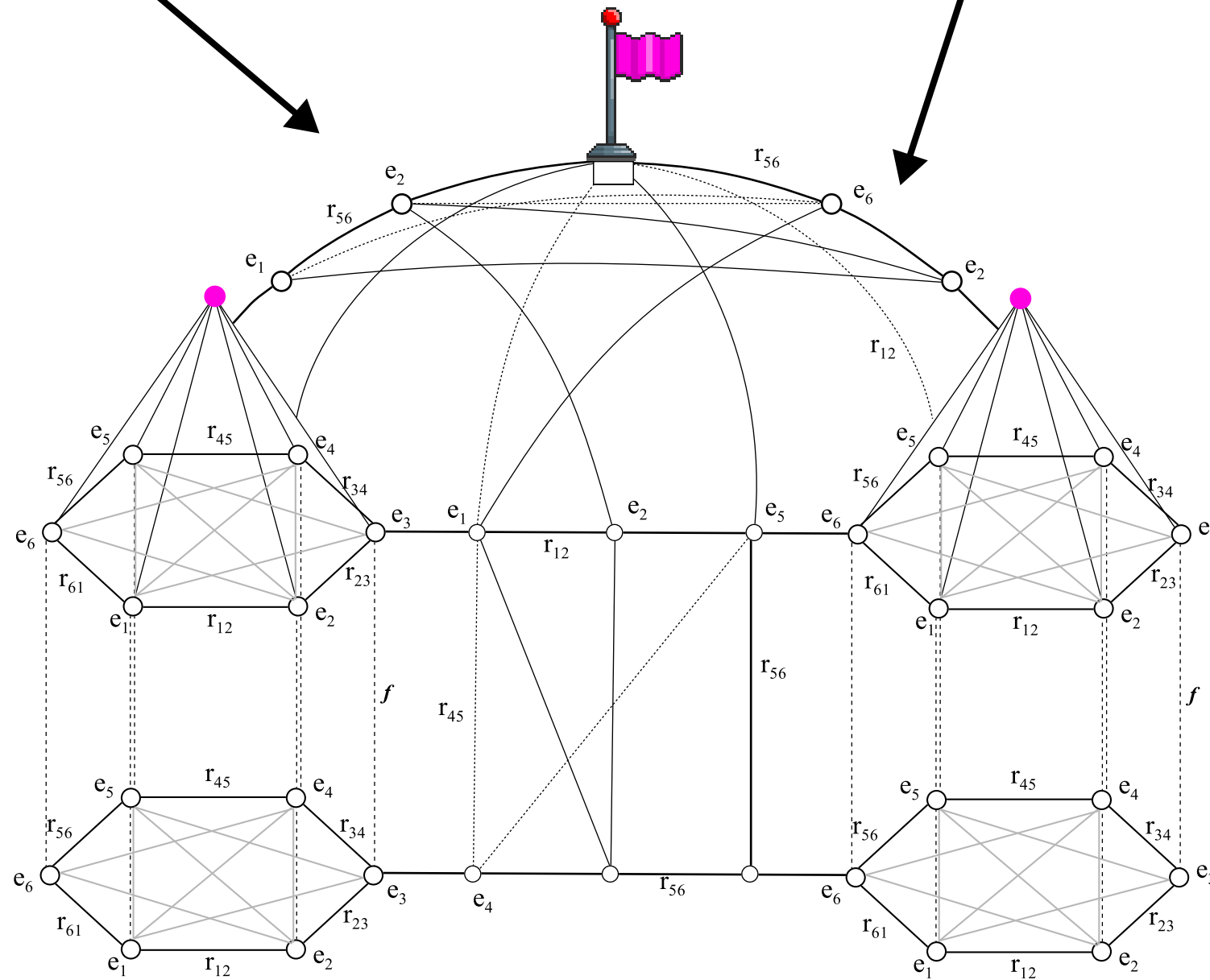




From [Altabaa et al.](#) , ICLR 2024.

[ RCAs ]

# [ Formal Languages ]



A sentence is only valid if it follows the grammar of the language.

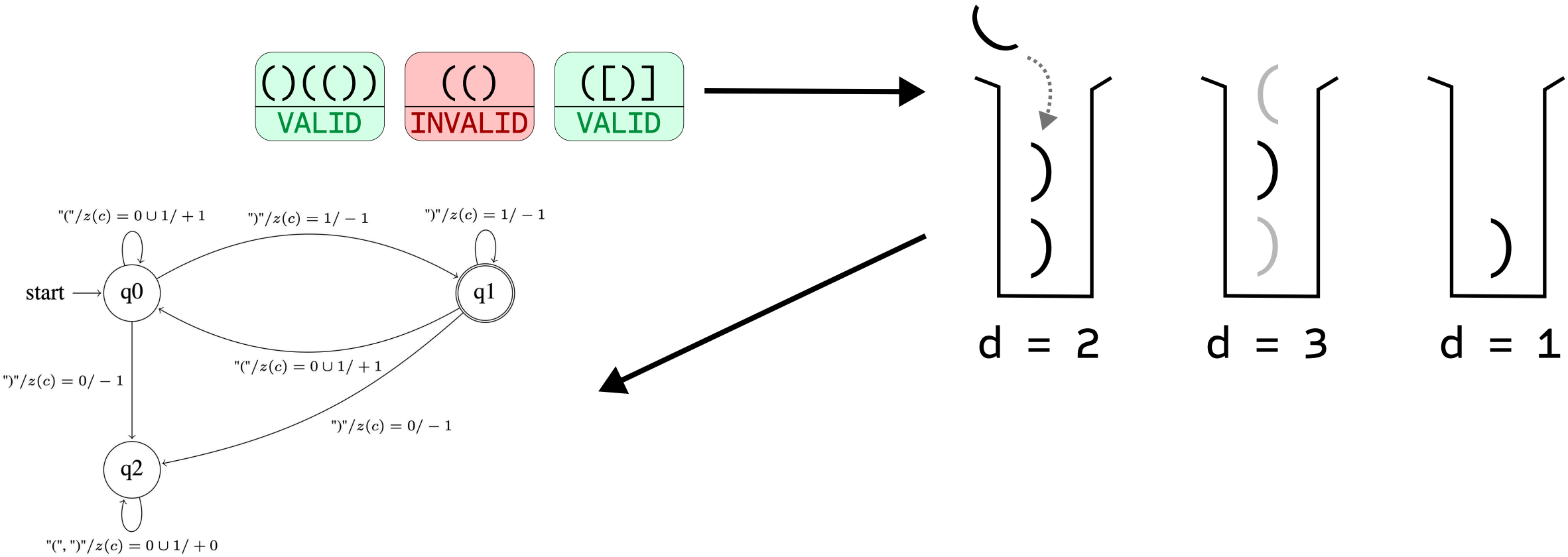
A string with  
letters from the  
alphabet  
  
the set of allowed  
symbols

member of the  
language

rules of  
concatenation,  
collapse,  
replacement, and  
conclusion

formal  
language

Counter Languages : All brackets in the string should be closed.



Our task can be translated to a **formal language**

Formal Languages can be defined by **automatons**

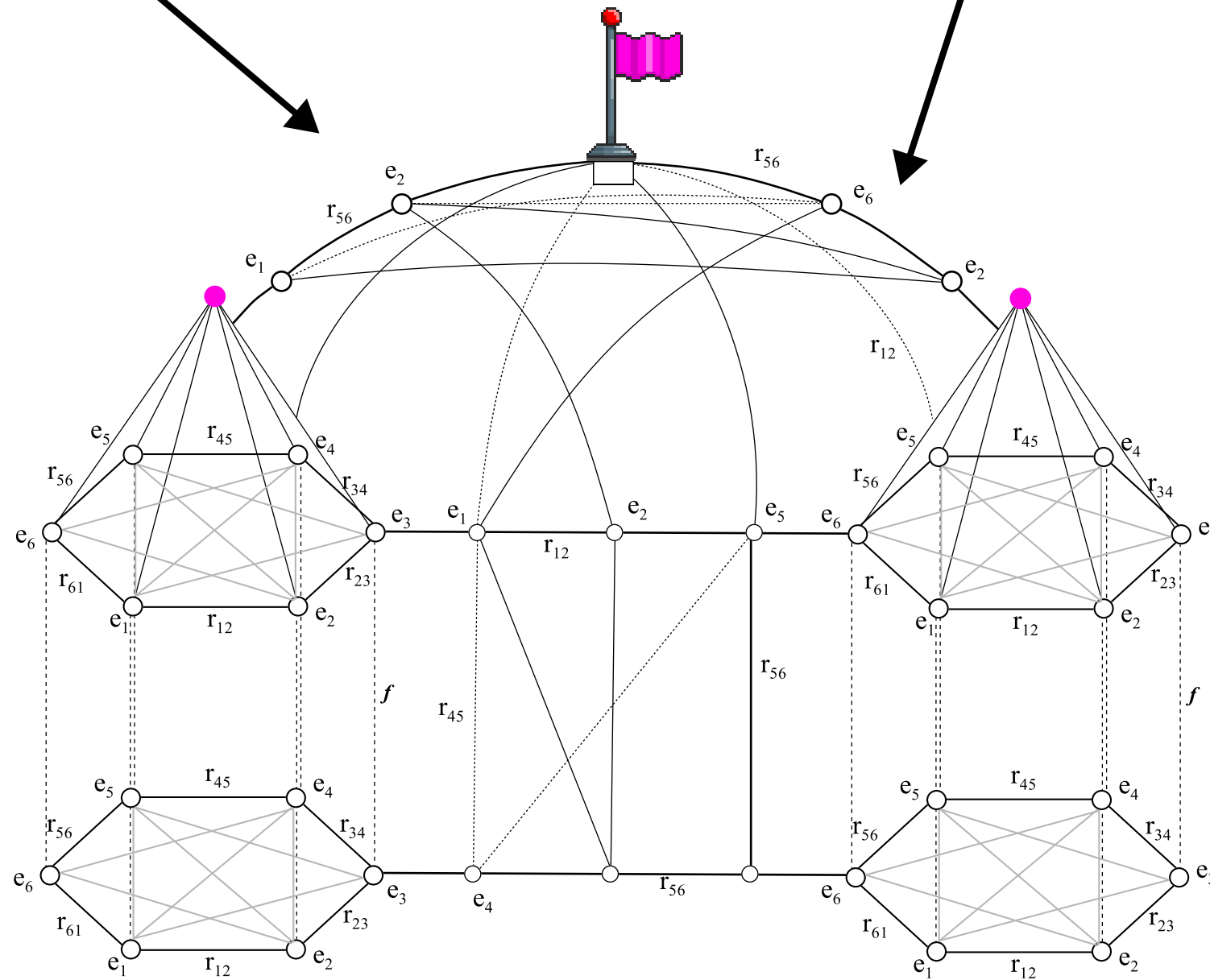
Automatons can be recovered **mechanistically\*** from  
transformers

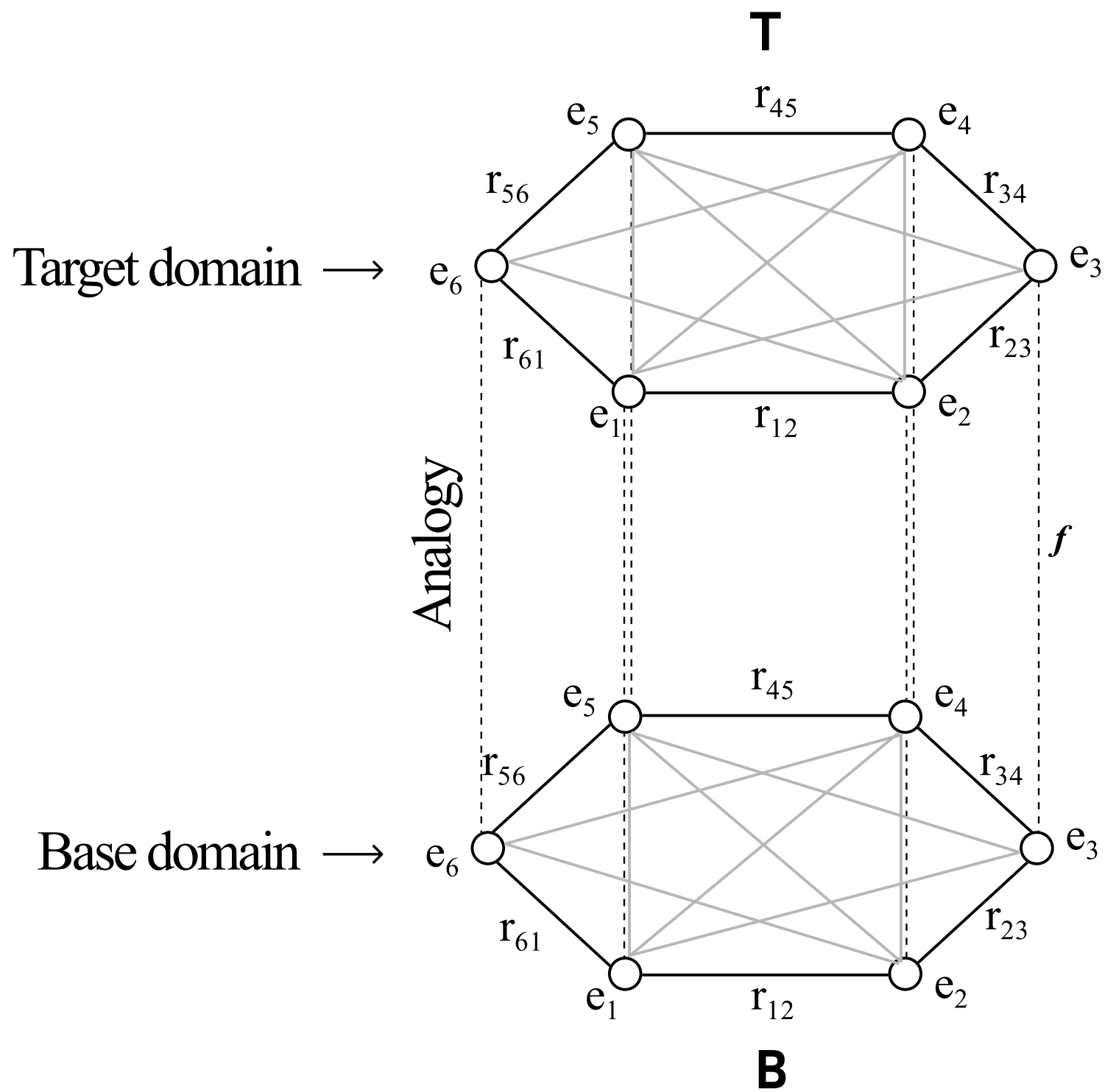
\* [Aliyah et al., 2024](#)

Why is it fickle?  
[ RCAs ]

What is happening?  
[ Formal Languages ]

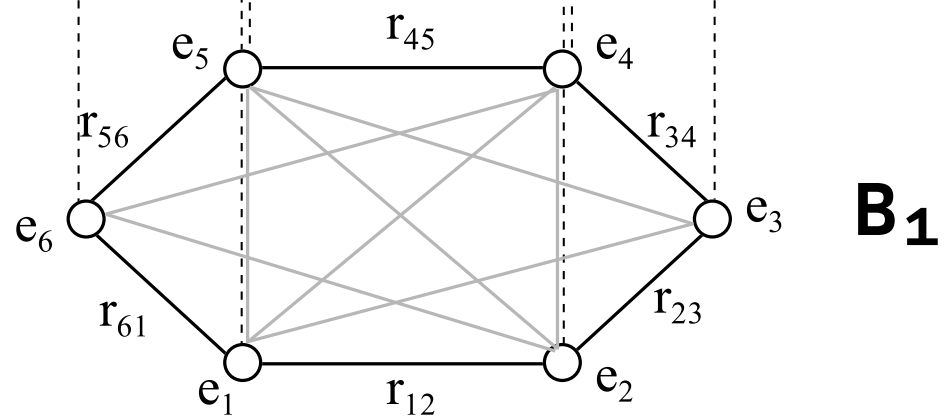
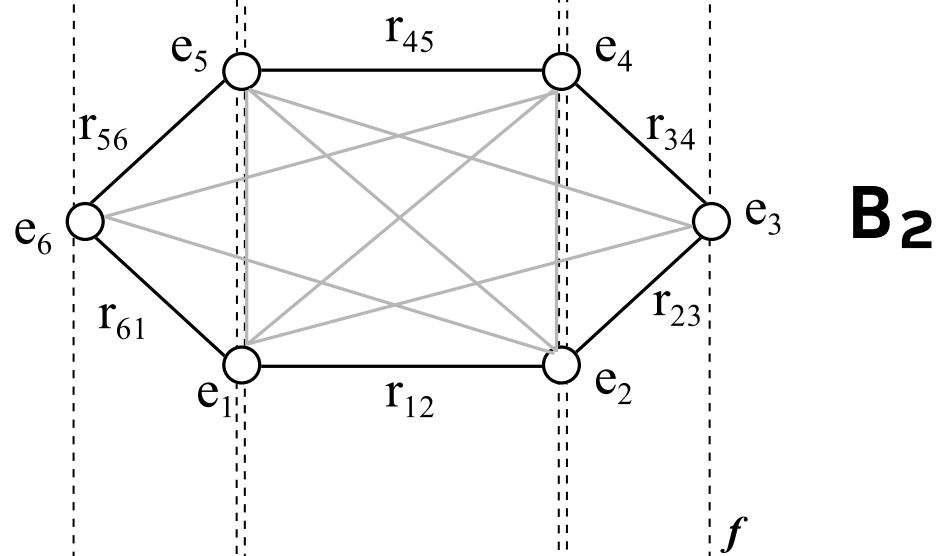
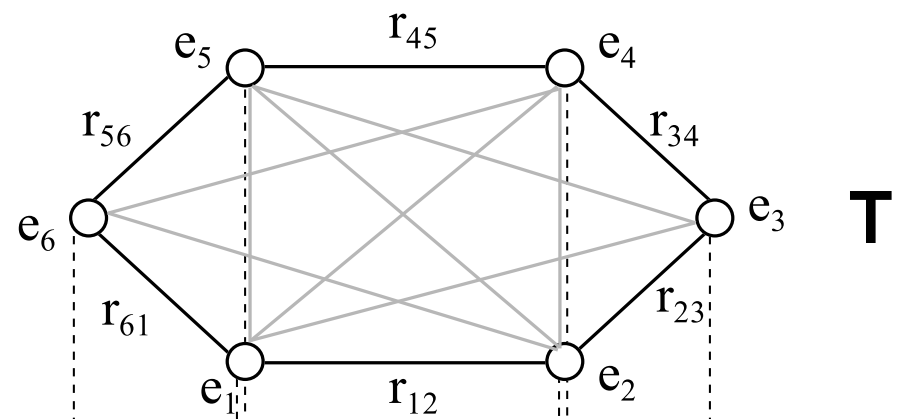
Whats the  
algebra of  
analogy?



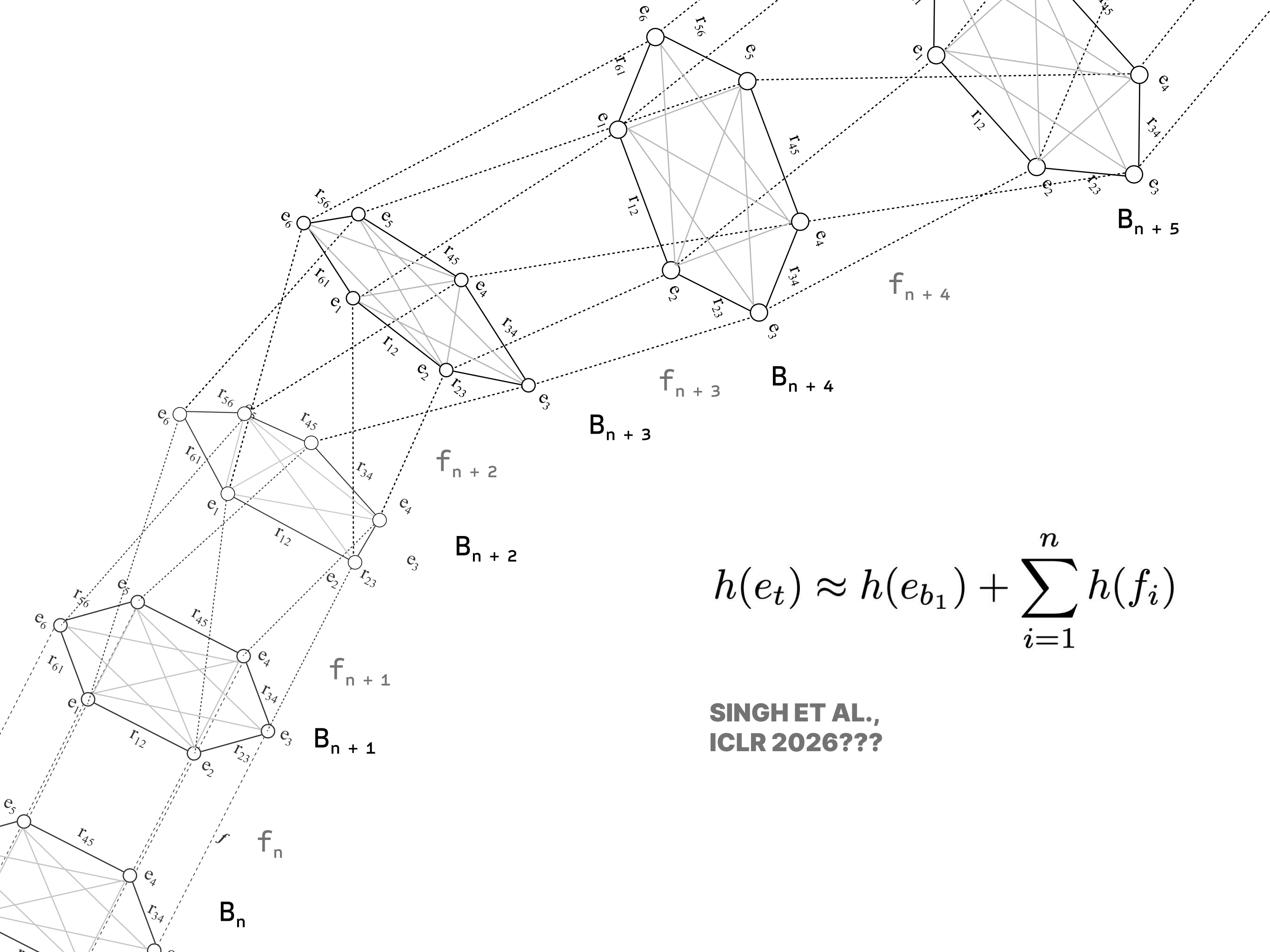


$$h(e_T) \sim h(e_B) + h(f)$$

MINIGESHI ET AL.,  
ICML 2026



$$h(e_T) \sim h(e_{B1}) + h(f_1) + h(f_2) ?$$



$$h(e_t) \approx h(e_{b_1}) + \sum_{i=1}^n h(f_i)$$

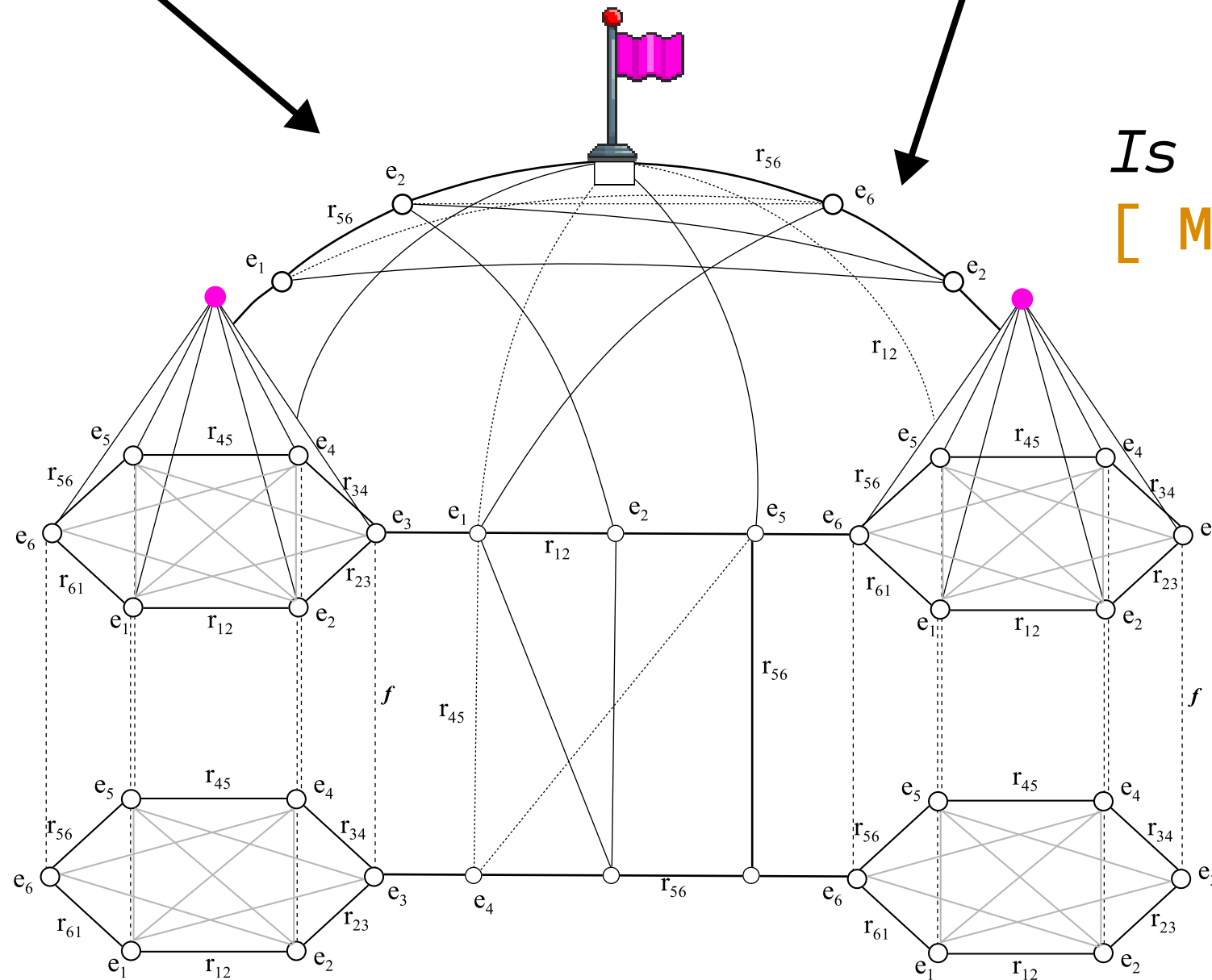
**SINGH ET AL.,  
ICLR 2026???**

Why is it fickle?  
[ RCAs ]

What is happening?  
[ Formal Languages ]

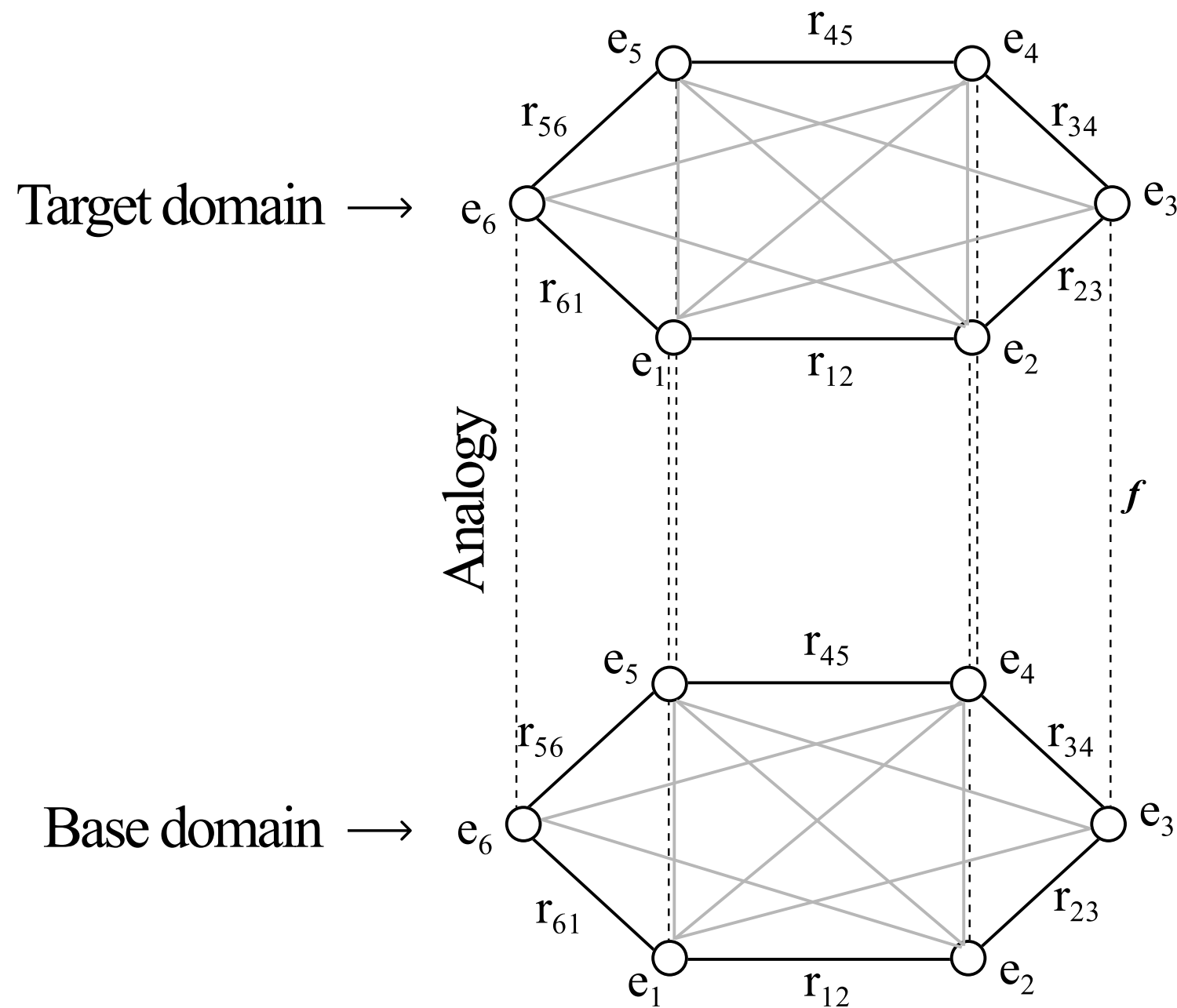
Is it pregnant?  
[ Meta-analogy ]

Whats the  
algebra of  
analogy?



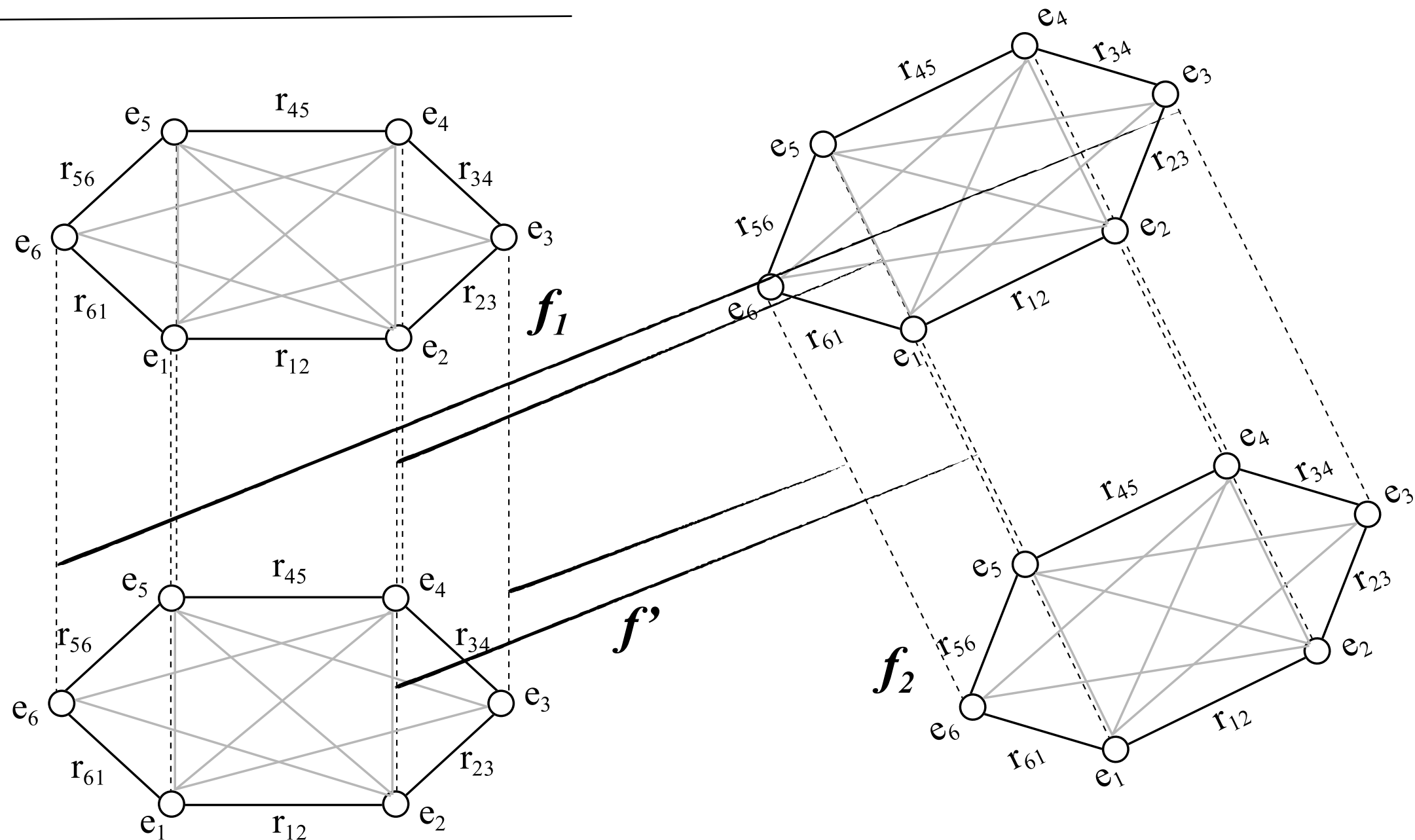
# Formalising meta-analogy, **METHOD 1**.

---



$$h(r_T) \sim h(r_B) + h(f) ?$$

# Formalising meta-analogy, **METHOD 2.**



$$h(f_2) \sim h(f_1) + h(f') ?$$

# Why I find it *so* interesting.

---

These functors between functors are super interesting, because they might compose. If they do, then that means these meta-functors form a category of embeddings – which are words.

That means this trail might lead us to words which are qualitatively “meta analogous” – kinda like how clicking the first hyperlink on any wikipedia article, and repetitively doing so. reaches the page philosophy in finite time. “the meta-est word”

$ty : D$